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GAMVA: A JAPANESE AUDIO-VISUAL MULTI-ANGLE SPEECH CORPUS

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ABSTRACT

Audio-visual speech recognition (AVSR) has contributed to improve Automatic Speech Recognition (ASR) accuracy in noisy environments. In real-world scenarios, a speaker does not always face to a camera. Therefore, it is obvious that a visual speech recognition (VSR) system should correctly recognize spoken contents not only from frontal but also non-frontal faces. In this paper, we introduce our efforts to build a new multi-angle audio-visual speech corpus: GAMVA. GAMVA consists of multi-angle lip images captured by 12 cameras, face feature points and speech sounds. So far GAMVA has 20 Japanese male subjects, and each subject uttered 25 types of Japanese daily-use greeting utterances. A baseline ASR, VSR and AVSR systems based on our previous works are built to evaluate our GAMVA data. Experimental results showed that we could recognize lip images at various angles and improve speech recognition accuracy in noisy environments.

Index Terms— multi-angle audio-visual database, audio-visual speech recognition, multi-angle visual speech recognition, noise robustness, GAMVA

1. INTRODUCTION

Recently, Automatic Speech Recognition (ASR) has been confirmed to have high recognition performance by using Deep Learning (DL), one of attractive artificial intelligence technology, and is used in various scenes such as voice input for mobile phones and car navigation systems. However, there is a problem that speech waveforms are degraded by audio noise in real environments, reducing the accuracy of speech recognition. In order to overcome this issue, we need to develop robust ASR systems against any audio noise. One of such ASR systems applicable in noisy environments is Audio Visual Speech Recognition (AVSR, also known as multimodal speech recognition), which employs ASR frameworks with Visual Speech Recognition (VSR, also known as lipreading). VSR uses lip images which are not affected by audio noise and estimates what a subject uttered only from a temporal sequence of lip images. VSR and AVSR have a potential to be applied in various practical applications such as automatic conference minute generation and human interface on smartphones. Owing to state-of-the-art DL technology, recently we have achieved high performance of VSR. However, VSR still has several problems when we try to use the technique in real-world scenes; for example, most of VSR research works have only consider frontal faces, but VSR technology for non-frontal views is also essential for real applications. In other words, assuming real scenes, a speaker does not always face to a camera such as smart device or tablet device on a VSR or an AVSR system.

In order to build VSR or AVSR systems, we need a large-size corpus having lip or face images. Currently, there are many databases available for the purpose. For example, SSSD database [1] has LF-ROI (Lower-Face Region-Of-Interest) and facial feature points. CENSREC-1-AV [2], OuluVS2 [3] and GRID [4] have utterances recorded in audio and visual modalities. As long as we know, the only public data set that has been captured simultaneously at multiple angles is OuluVS2. OuluVS2 has five-angle lip movies, however, to investigate pose-invariant VSR, we need much more views, not only horizontally but also vertically.

Therefore, in this paper, we introduce our efforts for "GAMVA (Gifu Audio-visual Multi-angle Videos with Annotation)" which is a new multi-angle audio-visual database recorded using 12 cameras. To build the database, we located 10 cameras horizontally every 10 degrees from frontal to profile, in addition to upper and lower cameras in front of a speaker. We started to collect utterance data from the simple classification task, as our first step and to confirm our recording equipment. Our corpus is designed to be used with the other Japanese corpus such as SSSD and CENSREC-1-AV. So far, we recorded utterances by 20 male subjects and composed facial feature points obtained from multi-angle video images, in addition to utterance speech waveforms and LF-ROI. Each speaker in our corpus uttered 25 Japanese greeting sentences, of which 15 greetings have been adopted in SSSD.

The rest of this paper is organized as follows. In Section

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Table 1. Speech corpora for lipreading and audio-visual speech recognition.

Database	Modality	#speakers	Contents	Language	Camera angles
OuluVS[5]	A, V	20	greeting phrases	English	frontal
OuluVS2[3]	A, V	52	digit sequences greeting phrases TIMIT sentences	English	0°, 30°, 45°, 60°, 90°
CUAVE[6]	A, V	36	digits	English	0°,90°
AVLetters[7]	A, V	10	isolated letters	English	frontal
AVICAR[8]	A, V	86	isolated digits isolated letters ten-digit phone numbers TIMIT sentences	English	four near frontal
GRID[4]	A, V	34	sentences	English	frontal
LRW[9]	A, V	1,000+	sentences	English	near frontal
M2TINIT[10]	A, V	1	ATR503 sentences	Japanese	frontal
CENSREC-1-AV[2]	A, V	93	consecutive digits	Japanese	frontal
SSSD[1]	V, F	48	isolated digits greeting phrases	Japanese	frontal
GAMVA	A, V, F	22	greeting phrases	Japanese	0°, 10°, 20°, 30°, 40° 50°, 60°, 70°, 80°, 90°, upper (0°), lower (0°)

Modality: A=Audio, V=Visual (face or lip), F=Feature points. #speakers represents the number of speakers.

2, we briefly review related open databases. Section 3 introduces our developing new corpus GAMVA. Section 4 introduces a baseline system. Experimental setup, results, and discussion are described in Section 5. Finally Section 6 concludes this paper.

2. RELATED WORKS

In this section, we briefly introduced existing corpora for VSR and AVSR. OuluVS [5] includes 20 speakers who uttered 10 phrases. As its next version, OuluVS2 has audio clips as well as multi-view lip images simultaneously recorded at the following five angles: 0°, 30°, 45°, 60° and 90°. This database includes 52 speakers, and each speaker uttered 10 digit sequences, 10 greeting phrases and 10 TIMIT sentences [11]. Most of multi-angle VSR or AVSR research works used this database. CUAVE [6] contains 36 subjects who uttered connected and isolated digits. AVLetters [7] has 10 speakers, and each person uttered isolated letters of English alphabet (A to Z) three times. AVICAR [8] includes 86 speakers uttering in a driving car. Each subject uttered four types of utterances: isolated digits, isolated letters, ten-digit phone numbers, and TIMIT sentences. GRID database contains 34 speakers each uttering sentences which consist of a small number of keywords. Regarding large-vocabulary AV speech corpora, The Oxford-BBC lipreading in the Wild (LRW) data set was introduced in [9]. The data set includes up to 1,000 utterances of 500 different words, spoken by more than 1,000 of different speakers.

Several Japanese speech recognition databases are available to the public. M2TINIT [10] is an audio-visual corpus which includes one male who uttered ATR-503 phonetically-balanced sentences CENSREC-1-AV is an audio-visual database for noisy environments, which contains 42 speakers for training and 51 speakers for testing. Each speaker uttered dozens of consecutive digit sentences. SSSD is a VSR corpus, which includes lip images and facial feature points from 48 speakers, recorded on mobile devices. Each speaker uttered 10 isolated numbers and 15 greeting phrases.

Details about the above databases (modality, the number of subjects, utterance contents, language, recording camera angles) are summarized in Table 1.

3. GAMVA

3.1. Utterances

To investigate multi-angle VSR and AVSR, we have collected audio-visual multi-view speech data. We named our new corpus GAMVA: Gifu Audio-visual Multi-angle Videos with Annotation. In the database, Japanese daily-use greeting phrase utterances were collected. The 25 Japanese utterances and corresponding English meanings are shown in Table 2. Utterances No.1–15 are the same as SSSD. SSSD employed additional 10 number utterances providing the 25-class classification task. As described later, since we have a plan to collect digit utterances as CENSREC-1-AV, this time we excluded number utterances. Instead, we chose additional 10 greeting utterances No.16–25. The subject uttered the ut-

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Table 2. Utterances in GAMVA.

No.	Japanese pronunciation	English meaning
1	/a-ri-ga-to-u/	thank you
2	/i-i-e/	no
3	/o-ha-yo-u/	good morning
4	/o-me-de-to-u/	congratulation
5	/o-ya-su-mi/	good night
6	/go-me-N-na-sa-i/	I'm sorry
7	/ko-N-ni-chi-wa/	good afternoon
8	/ko-N-ba-N-wa/	good evening
9	/sa-yo-u-na-ra/	good bye
10	/su-mi-ma-se-N/	excuse me
11	/do-u-i-ta-shi-ma-shi-te/	you are welcome
12	/ha-i/	yes
13	/ha-ji-me-ma-shi-te/	nice to meet you
14	/ma-ta-ne/	see you
15	/mo-shi-mo-shi/	hello
16	/ba-i-ba-i/	bye bye
17	/i-ta-da-ki-ma-su/	I'll take it
18	/go-chi-so-u-sa-ma/	thank you for meal
19	/o-tsu-ka-re-sa-ma/	see you
20	/ta-da-i-ma/	I'm home
21	/o-ka-e-ri/	welcome home
22	/o-jya-ma-shi-ma-su/	excuse me for disturbing you
23	/si-tsu-re-i-shi-ma-su/	excuse me
24	/hi-sa-shi-bu-ri/	long time no see
25	/yo-ro-shi-ku/	nice to meet you

terance set three times similar to OuluVS2. Note that, the speed of speech was not specified, and each speaker was asked to speak them as usual. So far, we collected data from 20 subjects: Japanese male university students in their early twenties.

3.2. Recording condition

Recording was conducted in office environments. We constructed a recording system for GAMVA. Fig.1 depicts horizontal camera locations in addition to a speaker. A subject sat on a green background, with a lapel microphone. As shown in Fig.1, 10 cameras were located every 10 degrees: 0° (frontal), 10°, 20°, $\cdots$, 80° and 90° (profile). Fig.2 illustrates vertical camera locations from the profile view. In addition to the frontal camera, we put additional two cameras. One camera was located at the lower level, while another was set at the upper level. All the cameras were roughly 90cm away from a speaker. These 12 cameras and the microphone simultaneously captured face movies and speech waveform. The sampling frequencies of speech and movie data frame rate were 16kHz and 30fps, respectively. The original movie size was 1280×720.

To record the data, we prepared four computers each which captured three cameras among them. Note that each camera movie contains not only face images but also speech

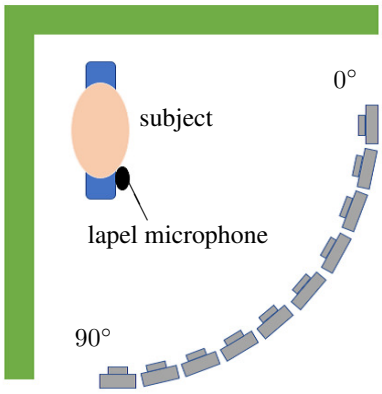


Fig. 1. Horizontal camera locations.

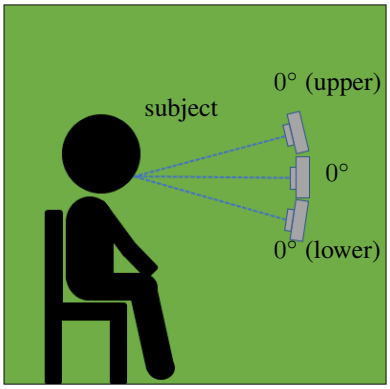


Fig. 2. Vertical camera locations.

signal. We also put another computer which recorded speech data and showed sentences to a speaker via LCD. In order to synchronize audio and visual data in post-processing and to inform a speaker to start uttering, we set a buzzer and a light on at the same time before and after each utterance.

3.3. Post-processing

At first, we extracted speech data by applying Voice Activity Detection (VAD). We used inaSpeech Segmenter [12] to carry out VAD, for waveforms not only recorded by microphone but also extracted from camera movies. After VAD, we synchronized all 13 waveforms for one utterance. Each audio data has approximately 500ms non-speech segments before and after the speech section.

Secondly, we extract lip images from every camera movies. We used FaceAlignment [13] to obtain facial feature points. The toolkit provides robust feature point extraction to multi-angle data from frontal to profile. Using the facial feature points, we finally extracted lip images. Fig.3 includes sample lip images.

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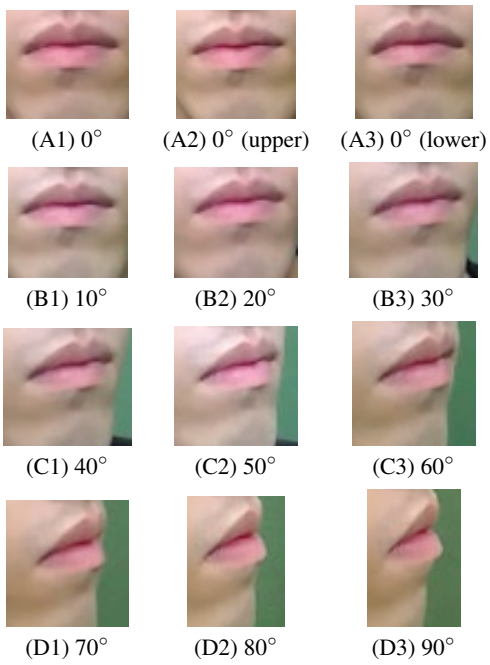


Fig. 3. An example of lip images.

4. A BASELINE SYSTEM

Using the audio-visual data introduced in the last section, We built a baseline classification system. Note that, in this paper "speech recognition" indicates not only phoneme-level speech recognition but also speech classification such as the baseline.

4.1. ASR

We developed a baseline method using GAMVA. For speech data, we adopted an ASR system using 2D Convolutional Neural Network (CNN) models. Details of the ASR model is shown in Fig.4. In our ASR framework, we extract mel-frequency spectrograms, which commonly used for CNN-based speech recognition. Since the length of utterance audio varies depending on utterances, the size of the mel-frequency spectrogram was fixed at 128×128. Note that when using the CNN (Conv2D), Batch Normalization (BN) was also applied.

4.2. Multi-angle VSR

We built multi-angle VSR scheme based on our previous work [14]; an angle estimator and angle-specific classifiers were prepared, and then, angle-specific results were integrated based on estimation results of the angle classifier. Because GAMVA also provides lip feature points, we modified the angle estimator to use the information.

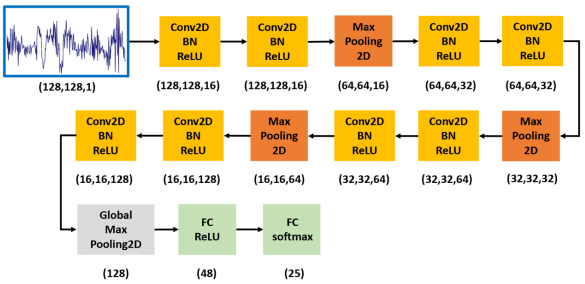


Fig. 4. An overview of our ASR model.

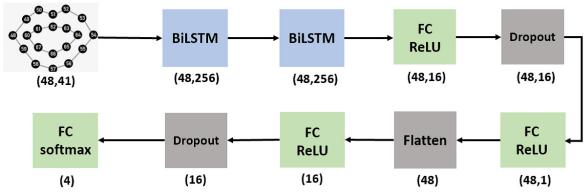


Fig. 5. An overview of angle estimator model.

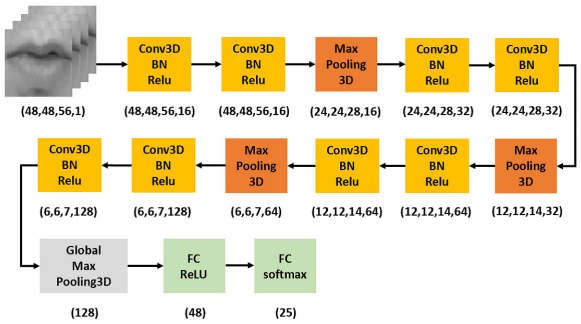


Fig. 6. An overview of angle-specific VSR model.

4.2.1. *Angle Estimator*

Assuming real environments, the angle of speaker face often varies and a VSR system should correctly deal with the phenomenon. One of the solution is to introduce an angle estimator to choose appropriate recognition model for the estimated view. The angle estimator model is shown in Fig.5. In this paper, because of the data limitation, we developed an angle estimator which classified an input image into four groups: (A) 0°, (B) 10°–30°, (C) 40°–60° and (D) 70°–90°. We chose a Bi-directional Long Short-Term Memory (BiLSTM) architecture in the model, followed by Fully-Connected (FC) layers.

We utilized coordinates of 20 low-face feature points. In addition to the x-axis and y-axis coordinates of each feature point, an x-axis difference between two horizontal ends of the outer lip contour was added. As a result, a 41-dimensional vector was obtained for an image. Similar to ASR, the length of each lip movie was not same. We then normalized the

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Table 3. Experimental data setup.

		Noise	#spkr	#data
Train	Audio	Clean	12	900
		Kitchen	12	900
		Meeting	12	900
		Station	12	900
	Visual	Clean	12	900 × 12 cameras
Valid	Audio	Clean	3	225
		Kitchen	3	225
		Meeting	3	225
		Station	3	225
	Visual	Clean	3	225 × 12 cameras
Test	Audio	Clean	5	375
		Park	5	375 × 5 SNRs
		Car	5	375 × 5 SNRs
	Visual	Clean	5	375 × 12 cameras

#spkr represents the number of speakers, and
#data represents the number of utterance data.

frame length to 48; if the original length was less than 48 we conduct upsampling, otherwise we suppress some frames.

4.2.2. Angle-specific model

We selected a 3DCNN-based model as an angle-specific VSR. Details of an architecture of our VSR system is illustrated in Fig.6. We prepared four models each corresponding to a group, either of (A)–(D). We normalized an image frame length to 48, similar to the angle estimator. Furthermore, the image size varied depending on speaker and angle. Thus we resized all images to 48×56. Before applying the CNN (Conv3D), we converted all lip images to gray-scale ones.

4.3. AVSR

We firstly carried out ASR and VSR separately. Since the task in this work was a 25-class classification, we then obtained probabilities for each class from ASR and VSR results respectively, by applying a softmax function. These probabilities were subsequently integrated in a decision fusion manner to obtain an AVSR score, by summing up ASR and VSR outputs. Finally the class having the highest score was selected as a classification result.

Let us denote conditional probabilities of class c from ASR and VSR models by $P_A(\mathbf{x}_A|c)$ and $P_V(\mathbf{x}_V|c)$, respectively. Here, $\mathbf{x}_A$ indicates an audio input representation, and $\mathbf{x}_V$ means the corresponding image vector. We then obtain an audio-visual probability $P_{AV}(\mathbf{x}_A, \mathbf{x}_V|c)$ as:

$$P_{AV}(\mathbf{x}_A, \mathbf{x}_V|c) = \alpha P_A(\mathbf{x}_A|c) + (1-\alpha)P_V(\mathbf{x}_V|c) \quad (1)$$

In this work, we simply fix $\alpha = 0.5$.

5. EXPERIMENT

5.1. Setup

We conducted speech recognition experiments to evaluate our new multi-angle audio-visual database and our baseline system. Table 3 shows details of data condition. In this paper, we performed 5-hold cross validation; among 15 speakers, we chose 12 subjects for training and the rest was used for validation. Note that the test set was fixed. We added acoustic noises in DEMAND [15] to original audio data. We chose three noise kinds for train and validation sets at different Signal-to-Noise Ratio (SNR) levels. We also employed different two noise kinds for testing.

5.2. Results

Table 4 shows multi-angle VSR results with or without angle estimation. The average classification accuracy of the angle estimator was 87.02%. We found that the presence or absence of the angle estimator had little effect on the recognition accuracy in the current setting. We also confirmed that the frontal angle was the highest, followed by 30°.

Table 5 shows the recognition accuracy of ASR, VSR and AVSR of the our baseline system. Focusing on the ASR results, the system achieved high recognition performance even when the SNR was low. This is due to the fact that noisy training data were used for training as augmentation data, and the task was the 25-class classification, which is simpler and easier than conventional speech recognition tasks such as connected-digit and large-vocabulary ones. The VSR accuracy was consistent and unrelated to SNR since visual information is not affected by audio noise. The VSR accuracy was lower than ASR performance in all the SNRs, because audio features are more effective and informative than visual ones. Finally, it is found that AVSR achieved the best accuracy in all the noise conditions. Similar to ensemble approaches, we believe our decision-fusion method could successfully integrate ASR and VSR results, which had different recognition errors.

6. CONCLUSION

In this paper, we introduced our efforts to build a new audio-visual database for multi-angle audio-visual speech recognition: GAMVA. GAMVA has multi-angle human lip images captured by 12 cameras, corresponding face feature points as well as speech waveforms. We then evaluated our data using a CNN-based baseline system.

Our future works include the increase of data variation; we will record not only male but also female subjects, from young to elder generations. In addition, we are planning to collect the other tasks, such as connected-digit and large-vocabulary data.

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Table 4. Recognition accuracy (%) of multi-angle VSR with or without angle estimation.

	0°	0°(l)	0°(u)	10°	20°	30°	40°	50°	60°	70°	80°	90°	Mean
Without AE	83.25	78.40	79.84	78.67	79.31	82.13	78.56	75.79	77.55	73.07	72.91	71.57	77.59
With AE	82.99	78.99	79.47	80.75	79.41	81.92	78.72	75.89	76.80	75.36	73.01	71.57	77.91

AE = Angle Estimator, 0°(l) = 0° from the lower camera, 0°(u) = 0° from the upper camera.

Table 5. ASR, VSR and AVSR recognition accuracy (%) in various noise conditions.

	clean	Park noise					Car noise				
		SNR0	SNR5	SNR10	SNR15	SNR20	SNR0	SNR5	SNR10	SNR15	SNR20
ASR	94.13	92.21	94.03	94.19	94.08	94.13	90.67	93.65	94.19	94.24	94.29
VSR	77.91										
AVSR	**97.88**	**97.33**	**97.75**	**97.80**	**97.86**	**97.86**	**96.88**	**97.61**	**97.80**	**97.84**	**97.88**

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